» Starting a job after my PhD

Dr. Marwa El Hefnawy, PhD in Electrical Engineering 2015: After successfully completing her PhD at TUM, Marwa El Hefnawy has already put her talents and knowledge to good use at several major tech firms. In this brief interview, she explains how she made the leap from academia to the private sector and what she has learned in the process.

TUM Alumna Dr. Marwa El Hefnawy obtained her Bachelor's degree from the German University in Cairo with summa cum laude honors and wrote her Bachelor's thesis at the University of Stuttgart. She won a scholarship from the German Academic Exchange Service (DAAD) and went on to do a Master's degree at Ulm University. At TUM, she completed her doctorate in new waveforms for 5G, before 5G was standardized, focusing specifically on faster-than-Nyquist (FTN) signals. After her PhD, she decided to find a job as a telecommunications engineer in the private sector, where she worked on 5G issues in her role as an experienced strategy planning manager for cellular modems and wireless system engineering. She brought her expertise to positions at tech firms such as Intel and Apple. Dr. Marwa El Hefnawy also contributes her time as a mentor in the *TUM Mentoring by Alumni for Students* and is a member of Women of TUM. She has spent many years as a global ambassador for the Society of Women Engineers (SWE) and is co-chair of the advisory board of the WE Local Europe conference.

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What made you decide to look for a job in industry after your PhD?

At university, you're doing research on lots of things that may not have practical use in the industry. I really wanted to develop something that had the potential to become reality. I wanted to learn how to actually deliver a real product to the customers, how to assess costs and resources correctly, and other surrounding factors, which go along with them.

Did you find it hard to find a suitable position in industry?

There were enough vacancies out there, but it took me some time before I found something that I felt was really interesting. I could have gone into software development, for instance. There were a lot of opportunities there, but it wasn't quite what I was after. I wanted to apply all my learnings and acquired skills during my undergrad and grad studies in my future job, be able to develop and innovate something and make a contribution to the 5G telecommunications technology. I then found a job in that line at one of the big international companies. I helped to develop a product that which people were genuinely able to hold in their hands in the end.

From your experience, what skills do employees value in PhDs in particular?

Well, there are certainly lots and lots of things involved in here. They are not just the competencies that are directly related to the job, such as coding knowledge for example, which are important. Soft skills are relevant too, such as the ability to present, sell your work and your product successfully and be an excellent team player or leader. Neither should you forget that life in industry is much more fast-paced nor should you underestimate the ongoing constant change. Therefore, flexibility and resilience are key attributes. For example, if you managed a project during your PhD where you had to act very flexibly at a short notice, this skill will undoubtedly be greatly appreciated.

How were your first 100 days on the job?

Starting out in a major international firm after my PhD was a challenge because I was the only lady in my team. Initially, it was hard to get my voice heard in meetings and engage my colleagues in conversations, but it got better when I started breaking the ice, joining the work flow and understanding the workspace culture. The longer I was at the company, I kept taking forward steps, and in the end we were a fantastic team. Also, however, it took me some time to realize that I had to change my way of thinking. From that point on, I had to think entrepreneurially, not just academically. In the academic world, there aren't really any barriers to research, and that's a good thing. However, in the industry, I suddenly found myself facing limits, such as costs or available resources or harder timelines or external factors in the market. I had to learn how to work with these in such a way that I was also capable of making decisions and achieving good results within a fixed framework.

Would you like to hear more?
TUM Alumna Marwa El Hefnawy talks about her experiences moving from academia to industry in the "That's my Job" careers podcast.
www.community.tum.de/en/podcast

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